Paul Karels

Staff / Senior Android Software Engineer

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EXECUTIVE SUMMARY

High-impact **Senior Android Engineer with 16 years of professional experience** specializing in native mobile architecture, robust offline-first synchronization, and secure enterprise applications. Proven track record of building production-grade mobile platforms from the ground up, modernizing legacy systems, and leading mobile initiatives within heavily regulated sectors, including **FinTech, Healthcare, and Enterprise EHSQ**. Expert in Kotlin, architecture components, high-throughput CI/CD automation, and low-level NDK/JNI implementations.

TECHNICAL SKILLS

Languages & Core:	Android SDK (Kotlin, Java), NDK/JNI, Coroutines, Flow, .NET Development
Architecture:	Modern Android Architecture (MVVM, MVI), Jetpack Compose, Dagger/Hilt, Jetpack Components
Data & Cloud:	Room, Offline Data Sync, Background WorkManager, OkHttp, Retrofit, REST APIs, GCP (Certified)
DevOps & Security:	Google Cloud Build, Firebase Test Lab, Docker, Veracode Security Scanning, ProGuard

PROFESSIONAL EXPERIENCE

Ideagen | Senior Software Engineer (Mobile Lead) 04/2023 – Present

- **Architecture Ownership:** Serve as the primary Android engineer owning the end-to-end architecture, security, and production stability of the native mobile app for Ideagen DevonWay, an enterprise platform used in regulated industries.
- **Greenfield Engineering:** Architected and engineered the company's flagship native Android app completely from scratch in Kotlin, successfully scaling it through release 2.6.
- **Low-Level Integration:** Spearheaded the integration of a core shared C forms engine using JNI/NDK, building 20+ dynamic field types including digital signatures, interactive maps, and transactional attachments.
- **Resilient Systems:** Designed an offline-first data sync framework and background form-submission system with automated retry/replay logic, preserving data integrity across highly volatile field networks.
- **CI/CD Automation:** Automated the entire distribution pipeline using Google Cloud Build, integrating automated unit/UI tests (Firebase Test Lab), security compliance scanning (Veracode), and seamless Play Store deployments.
- **Technical Leadership:** Mentor and code-review junior developers, establishing clean code patterns and architecture guidelines to ensure codebase health.

Nerdery | Senior Software Engineer (Mobile Consultant) 10/2018 – 04/2023

- Delivered high-quality, production-ready applications for enterprise clients across diverse industries, focusing on rapid time-to-market and robust mobile architecture.
- **Client Spotlight: National Financial Services Firm (Android Engineer)**
  - Modernized a native financial application used by retail clients to manage highly sensitive account data and banking details.
  - **Performance Engineering:** Drove user satisfaction metrics up significantly, raising the Google Play Store app rating

from under 4.0 to a stable 4.5 through performance optimizations and deep memory leak diagnostics.
• **Secure Feature Delivery:** Seamlessly integrated 3rd-party SDKs to deploy high-value features, including a secure biometrics/login workflow and a customized customer loyalty system.

- **Client Spotlight: Regional Medical Provider (Android Engineer)**
  • Co-designed and developed an agile, native Android application helping healthcare providers rapidly locate and process patient referrals.
  • Established the application's initial CI/CD automation pipeline to reduce delivery friction and accelerate developmental iteration cycles.
- **Client Spotlight: Materials Testing Firm (.NET Engineer)**
  • Engineered a high-performance Universal Windows Platform (UWP) desktop application leveraging SignalR to facilitate real-time bidirectional server communication with proprietary testing hardware.

Magenic Technologies | Consultant & Technical Instructor 2010 – 08/2018

- Provided high-end technical consulting and software delivery for Fortune 500 clients across multiple tech stacks, while running internal technical training seminars for mid-level and senior developers.
- **Client Spotlight: National Financial Investment Services Firm (Android Developer)**
  • Engineered native mobile modules handling critical financial transactions, allowing customers to view live account quotes and safely execute high-volume stock option trades.
- **Client Spotlight: National Financial Services Firm (Android Developer)**
  • Successfully executed a complete architectural pivot, converting a legacy PhoneGap hybrid application into a high-performance, native Android application, dramatically improving transaction speeds.
- **Client Spotlight: National Insurance Services Firm (Android Developer)**
  • Led the UI and backend logic redesign of the client's core insurance claims processing flow, significantly flattening the customer submission funnel.
- **Client Spotlight: National Commercial Air Handling Firm (Mobile Developer)**
  • Architected a rich, offline-capable B2B sales tablet application utilizing Xamarin.Android and Azure Mobile Services to synchronize complex data structures seamlessly.

EDUCATION & CERTIFICATIONS

- **Bachelor of Science (B.S.) in Computer Science** – North Dakota State University
- **Certified Associate Cloud Engineer** – Google Cloud Platform